



SCHOOL OF APPLIED SCIENCE
MASTER OF COMPUTER APPLICATIONS

Mini Project

Synopsis

For

Online Assessment System

*Submitted in partial fulfilment of the requirement for the award of the Degree
of*

Master of Computer Applications

By

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Introduction

With the rapid growth of digital education, online assessment systems have become an essential part of modern learning environments. Traditional examination methods such as paper-based tests are time-consuming, resource-intensive, and prone to human errors during evaluation. Educational institutions require a secure, scalable, and efficient digital platform to conduct assessments remotely while ensuring fairness and accuracy.

An Online Assessment System using Python Flask provides a web-based solution where administrators can create tests, students can attempt exams online, and results can be evaluated automatically. This system improves accessibility, reduces manual workload, and ensures quick result generation.

Problem Statement

Conventional examination systems face several challenges such as manual question paper preparation, delayed result processing, lack of transparency, and difficulty in managing large numbers of students. During remote learning scenarios, conducting secure and reliable assessments becomes even more difficult without a centralized online platform. Hence, there is a need for a web-based assessment system that supports online exams with automated evaluation and result management.

Objectives

The main objectives of the Online Assessment System are:

- To develop a secure and user-friendly online examination platform
 - To allow administrators to create, manage, and schedule assessments
 - To enable students to attempt exams online within a fixed time limit
 - To automatically evaluate objective-type questions
 - To generate instant results and performance reports
 - To reduce paperwork and manual evaluation efforts
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System Overview

The Online Assessment System follows a client-server architecture.

- The **frontend** provides interfaces for students and administrators to interact with the system.

- The **backend**, built using Python Flask, handles authentication, question management, exam logic, and result evaluation.
- The **database** stores user details, questions, exam records, and results securely.

The system supports role-based access, ensuring that only administrators can create or manage tests, while students can only attempt assigned exams.

Technologies Used

- **Frontend:** HTML, CSS, JavaScript
 - **Backend:** Python Flask
 - **Database:** SQLite / MySQL
 - **Authentication:** Session-based login system
 - **Tools:** VS Code, Browser-based testing
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System Workflow

- The administrator logs into the system and creates an assessment by adding questions and setting time limits.
 - Students register and log in using valid credentials.
 - During the exam, students attempt multiple-choice questions within the allotted time.
 - Answers are submitted to the Flask backend for validation.
 - The system automatically evaluates responses and stores results in the database.
 - Students can view their scores, and administrators can analyze performance reports.
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Limitations of the Earlier Version

- Supports mainly objective-type questions (MCQs)
 - No AI-based proctoring or cheating detection
 - Limited scalability for very large user loads
 - Requires stable internet connectivity
 - Basic UI design without advanced animations
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Conclusion

The Online Assessment System using Python Flask provides an efficient and reliable solution for conducting digital examinations. It minimizes manual intervention, ensures faster evaluation, and improves accessibility for both students and administrators. Although basic

in scope, the system lays a strong foundation for future enhancements such as intelligent monitoring, analytics, and large-scale deployment in educational institutions.